

Addendum for DPP 2016 August West Area Group Study

June 17, 2019

This addendum contains updated study results for the Report entitled *MISO DPP 2016 August West Area Phase 3 Study* and dated March 14, 2019. Results have been updated to reflect the Ottumwa transformer rating correction and its subsequent Network Upgrade changes.

The Ottumwa transformer rating correction has two major changes:

- 1) The correct ratings for Ottumwa 345-161 kV transformer are **CEII** MVA SN/SE. With these correct ratings, Ottumwa 345-161 kV transformer is no longer a constraint. Therefore, the original Network Upgrade (add 2nd 450 MVA transformer, \$9 M) is no longer required.
- 2) Without Ottumwa transformer Network Upgrade (2nd Ottumwa transformer), Parnell-Hills 161 line is no longer a constraint. Therefore, the original Network Upgrade (add 1 stage of 36 MVAR nominal, 27.6 MVAR effective cap bank at Parnell 161 kV, \$1.4 M) is no longer required.

The revised per project cost summary of generation projects is listed below.

1.1.1 J302 Summary

Network Upgrade	Cost	J302	NUs Type
J302&J503 POI-Heskett 230 kV	\$9,000,000	\$4,554,000	MISO SH & MDU LPC
Hankinson-Forman 230 kV	\$650,000	\$328,900	MISO SH
Oakes-Forman 230 kV	\$19,950,000	\$10,094,700	MISO SH
Oakes-Ellendale 230 kV	\$20,500,000	\$10,373,000	MISO SH
Merricourt-Ellendale 230 kV	\$4,600,000	\$2,327,600	MDU LPC
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$382,067	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$489,580	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$134,627	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$233,193	Reactive Power NU
Square Butte-Stanton 230 kV (NRIS)	\$10,975,000	\$5,553,350	NRIS
Merricourt-Ellendale 230 kV (NRIS)	\$50,000	\$25,300	NRIS
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$1,124	NRIS
J302 POI-J607 POI-Wishek 230 kV (NRIS)	\$750,000	\$379,500	NRIS

Network Upgrade	Cost	J302	NUs Type
Wishek-Merricourt 230 kV (NRIS)	\$850,000	\$430,100	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$35,307,041	

1.1.2 J476 Summary

Network Upgrade	Cost	J476	NUs Type
J530 POI-Hills 345 kV	\$27,000,000	\$1,560,562	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$624,846	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$463,179	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$928,975	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$89,520	Reactive Power NU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$6,489	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,673,571	

1.1.3 J503 Summary

Network Upgrade	Cost	J503	NUs Type
J302&J503 POI-Heskett 230 kV	\$9,000,000	\$4,446,000	MISO SH & MDU LPC
Hankinson-Forman 230 kV	\$650,000	\$321,100	MISO SH
Oakes-Forman 230 kV	\$19,950,000	\$9,855,300	MISO SH
Oakes-Ellendale 230 kV	\$20,500,000	\$10,127,000	MISO SH
Merricourt-Ellendale 230 kV	\$4,600,000	\$2,272,400	MDU LPC
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$373,007	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$477,969	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$131,434	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$227,663	Reactive Power NU
Square Butte-Stanton 230 kV (NRIS)	\$10,975,000	\$5,421,650	NRIS
Merricourt-Ellendale 230 kV (NRIS)	\$50,000	\$24,700	NRIS
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$1,098	NRIS
J302 POI-J607 POI-Wishek 230 kV (NRIS)	\$750,000	\$370,500	NRIS
Wishek-Merricourt 230 kV (NRIS)	\$850,000	\$419,900	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$34,469,720	

1.1.4 J512 Summary

Network Upgrade	Cost	J512	NUs Type
Helena-Scott Co 345 kV	\$54,000,000	\$22,404,560	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$1,457,618	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$1,354,958	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$430,170	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$386,416	Reactive Power NU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$3,384	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,037,106	

1.1.5 J541 Summary

Network Upgrade	Cost	J541	NUs Type
J530 POI-Montezuma 345 kV	\$350,000	\$157,374	MISO SH
J530 POI-Hills 345 kV	\$27,000,000	\$10,342,658	MISO SH
Zachary 345/161 kV transformer	\$7,000,000	\$4,000,000	Ameren LPC
Zachary-Adair 161 kV line	\$2,000,000	\$1,142,857	Ameren LPC
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$5,164,988	Reactive Power NU
Novelty 161 -69 kV xfmr	\$3,270,000	\$3,270,000	AECI AFS
South River-Emerson 161 kV	\$110,000	\$110,000	AECI AFS
J530 POI-Hills 345 kV (SNU)	\$300,000	\$56,466	SNU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$33,403	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$24,277,746	

1.1.6 J555 Summary

Network Upgrade	Cost	J555	NUs Type
J530 POI-Montezuma 345 kV	\$350,000	\$74,581	MISO SH
J530 POI-Hills 345 kV	\$27,000,000	\$4,915,735	MISO SH
Parnell-J438 POI 161 kV	\$250,000	\$250,000	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$442,848	Reactive Power NU

Network Upgrade	Cost	J555	NUs Type
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$371,523	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$2,198,028	Reactive Power NU
J530 POI-Hills 345 kV (SNU)	\$300,000	\$53,463	SNU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$14,132	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,320,310	

1.1.7 J569 Summary

Network Upgrade	Cost	J569	NUs Type
Helena-Scott Co 345 kV	\$54,000,000	\$7,579,499	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$553,316	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$510,070	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$192,851	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$141,295	Reactive Power NU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$1,483	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,978,515	

1.1.8 J583 Summary

Network Upgrade	Cost	J583	NUs Type
J530 POI-Hills 345 kV	\$27,000,000	\$1,755,753	MISO SH
Grimes-Sycamore 345 kV #2	\$2,200,000	\$2,200,000	MISO SH
Bondurant-Sycamore 345 kV	\$1,000,000	\$1,000,000	MISO SH
Bondurant-Montezuma 345 kV	\$200,000	\$600,000	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$652,054	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$533,378	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$1,082,081	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$58,562	Reactive Power NU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$7,320	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$7,889,147	

1.1.9 J587 Summary

Network Upgrade	Cost	J587	NUs Type
Helena-Scott Co 345 kV	\$54,000,000	\$16,799,157	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$886,888	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$1,047,667	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$280,020	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$818,808	Reactive Power NU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$2,309	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$19,834,848	

1.1.10 J590 Summary

Network Upgrade	Cost	J590	NUs Type
Helena-Scott Co 345 kV	\$54,000,000	\$7,216,784	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$765,604	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$539,157	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$188,109	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$9,148	Reactive Power NU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$1,427	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$8,720,230	

1.1.11 J598 Summary

Network Upgrade	Cost	J598	NUs Type
J530 POI-Montezuma 345 kV	\$350,000	\$118,045	MISO SH
J530 POI-Hills 345 kV	\$27,000,000	\$7,758,196	MISO SH
Zachary 345/161 kV transformer	\$7,000,000	\$3,000,000	Ameren LPC
Zachary-Adair 161 kV line	\$2,000,000	\$857,143	Ameren LPC
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$3,874,260	Reactive Power NU
J530 POI-Hills 345 kV (SNU)	\$300,000	\$54,429	SNU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$25,059	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$15,687,133	

1.1.12 J611 Summary

Network Upgrade	Cost	J611	NUs Type
J530 POI-Hills 345 kV	\$27,000,000	\$667,095	MISO SH
J611-Maryville 161 kV	\$1,000,000	\$1,000,000	MISO SH
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$247,461	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$175,293	Reactive Power NU
2x150 Mvar switched cap bank at Hills 345 kV (636400)	\$15,000,000	\$394,458	Reactive Power NU
1×50 Mvar switched cap bank at McLeod 230 kV (619940)	\$2,000,000	\$35,395	Reactive Power NU
J274 POI-Creston 161 kV (SNU)	\$160,000	\$54,990	SNU
Clarinda-Brooks 161 kV (SNU)	\$68,000	\$15,660	SNU
Montezuma-J530 POI 345 kV (NRIS)	\$100,000	\$2,773	NRIS
MCKSBRG-Winterset 161 kV (NRIS)	\$200,000	\$200,000	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,793,125	

1.1.13 J614 Summary

Network Upgrade	Cost	J614	NUs Type
Harmony-Cresco 69 kV	\$4,000,000	\$4,000,000	MISO SH & DPC LPC
2×75 Mvar switched cap bank at Killdeer 345 kV (631199)	\$6,500,000	\$114,291	Reactive Power NU
2×75 Mvar switched cap bank at Hickory Creek 345 kV (631191)	\$6,500,000	\$537,225	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,651,516	